MOHANGRAPHY

Photography Portfolio Website

Technical Guide & User Manual

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- **System Overview** — what the site is, all files at a glance
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- **Travel Stories Blog** — how to write articles, format body text, add sub-headings with ##, insert inline photos with [photo: filename.jpg]
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1. System Overview

Mohangraphy is your personal photography portfolio at mohangraphy.com. The entire site is generated by a single Python script on your Mac and published to the internet automatically. There is no CMS, no admin login, and no server to manage.

How it works in simple terms:

- You add photos to your Mac in the Photos folder
- You run curator.py to tag each photo with category, location and caption
- You run Claude_mohangraphy.py — it builds the entire website and pushes it live
- The site is live at mohangraphy.com within 30 seconds

1.1 Technology Used

Component	Technology
Site Generation	Python 3 — Claude_mohangraphy.py
Hosting	GitHub Pages (free, unlimited)
Database	Supabase — stores likes, visits, subscribers
Email Notifications	Resend — sends emails to subscribers when new photos are added
Analytics	Google Analytics GA4 + Plausible.io
Fonts	Cormorant Garamond (titles) + Montserrat (body text)
Domain	GoDaddy — mohangraphy.com
Automated Emails	GitHub Actions — fires automatically on every deploy

1.2 All Important Files at a Glance

File	Purpose
Scripts/Claude_mohangraphy.py	Main script — builds and publishes the entire website
Scripts/curator.py	Photo tagging tool — run this before deploying new photos
Scripts/blog_editor.py	Blog post editor — create and edit Travel Stories
Scripts/content.json	Site settings — email, analytics IDs, credentials
Scripts/photo_metadata.json	All photo tags, categories, locations, captions
Scripts/blog_posts.json	All Travel Stories blog articles
Scripts/list_subscribers.py	Admin tool — exports subscriber list to Excel
Photos/	Your original photos — never touched by any script
Thumbs/	Auto-generated 800px thumbnails — safe to delete anytime
Web/	Auto-generated 2048px web copies — safe to delete anytime
.github/workflows/notify.yml	GitHub Action — sends subscriber emails on every deploy

⚠️ **Never edit index.html directly. It is completely overwritten every time you run Claude_mohangraphy.py.**

2. Your Day-to-Day Workflow

Every time you want to add new photos to the website, follow these steps in order.

2.1 Adding New Photos — Step by Step

1	Copy your photos into the correct folder inside Photos/ on your Mac
2	Open Terminal (press Cmd+Space, type Terminal, press Enter)
3	Run the curator to tag your photos:

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/curator.py
```

4	A browser window opens — photo on the left, tagging form on the right
5	Select categories, fill in location details, add remarks if needed
6	Click Save & Next — it moves to the next photo automatically
7	When done, click Stop
8	Now run the deploy script to build and publish the site:

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

9	The script builds the site, pushes to GitHub, and prints "Live in ~30 seconds"
10	Done — your photos are live at mohangraphy.com

✅ **Subscribers receive an email automatically every time you deploy — no extra step needed.**

2.2 Recommended Folder Structure for Photos

Organise your photos by place — this makes it easy to find them and the tagging system works perfectly with this structure:

```
Photos/
  Places/
    National/
      Tamil Nadu/
        Megamalai/
        Madurai/
      Karnataka/
        Badami/
        Hampi/
      Kerala/
        Munnar/
    International/
      Canada/
        Banff/
  Nature/
    Landscapes/
    Wildlife/
    Birds/
    Flora/
```

```
Architecture/
People & Culture/
```

A single photo can belong to multiple categories. A bird photo taken in Megamalai can be tagged as both Places/National AND Nature/Birds — it will appear in both sections of the website.

2.3 Moving Photos to Different Folders

It is completely safe to reorganise your Photos folder at any time. The script scans all subfolders automatically.

Action	Result
Move photos into subfolders within same category	✅ Safe — redeploy after
Move photos from one category to another	⚠️ Safe, but re-tag via admin editor on website
Rename a folder	✅ Safe — redeploy after
Move photos outside the Photos/ folder	❌ Not safe — photos will disappear from site

✅ **After moving photos, always run Claude_mohangraphy.py. The script automatically cleans up old references from photo_metadata.json.**

3. Photo Tagging — curator.py

The curator is a browser-based tool that lets you tag your photos quickly. The photo appears on the left and the tagging form on the right — everything in one window, no switching between apps.

3.1 How to Run

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/curator.py
```

A browser window opens automatically at http://localhost:9797. The first photo in your Photos folder is shown immediately.

3.2 The Tagging Screen

Field	What to enter
Photo (left side)	Full photo shown — you can see exactly what you are tagging
Status badge	Shows NEW PHOTO (green) or ALREADY TAGGED (gold)
Progress counter	Shows "3 / 97" — which photo you are on out of the total
Categories	Click buttons to toggle on/off — gold = selected
State / Country	For Indian locations: enter the State (e.g. Tamil Nadu). For international: enter Country (e.g. Canada)
City	The specific place (e.g. Megamalai or Banff)
Remarks	Optional caption shown as overlay on the photo on the website
Date Added	Defaults to today — or carries forward from previous photo

3.3 Keyboard Shortcuts

Key	Action
Cmd + S	Save current photo and move to next
S	Skip current photo without saving

3.4 Carry-Forward Feature

When you are tagging a batch of photos from the same place, most fields stay the same. The curator remembers your previous entries:

- Categories → carried forward from previous photo
- State / Country → carried forward
- City → carried forward
- Date Added → carried forward
- **Remarks → always blank** — this is unique to each photo

When you move to a new location, simply type the new State/City values and they will carry forward from that point.

3.5 Categories Available

Category	Use for
Nature/Landscapes	Mountains, valleys, rivers, scenery
Nature/Wildlife	Animals, reptiles, insects
Nature/Birds	All bird photography
Nature/Flora	Flowers, plants, trees
Places/National	Indian locations — requires State + City
Places/International	Overseas locations — requires Country + City
Architecture	Temples, buildings, monuments
People & Culture	Portraits, street scenes, cultural events

Sunsets are included under Nature/Landscapes — there is no separate Sunsets category.

3.6 What Gets Saved

Every time you click Save, the details are written immediately to photo_metadata.json. You do not need to do anything extra — it is saved photo by photo as you go.

3.7 The photo_metadata.json File

This file stores all your photo information. Here is what a single entry looks like:

```
{
  "abc123hash": {
    "path": "Photos/Places/National/Tamil Nadu/Megamalai/bird.jpg",
    "categories": ["Places/National", "Nature/Birds"],
    "state": "Tamil Nadu",
    "city": "Megamalai",
    "country": "",
    "remarks": "Great Hornbill in flight",
    "date_added": "2026-04-01"
  }
}
```

⚠️ **Do not edit photo_metadata.json by hand unless you are confident with JSON. A single missing comma or quote will break the entire file.**

4. Website Navigation Structure

The website is organised into four main collections. Here is exactly how a visitor navigates through the site:

4.1 Collections Menu

Clicking Collections in the header shows four main categories:

Category	What you see next
Nature	Sub-categories: Landscapes · Wildlife · Birds · Flora
Places	Two cards: National and International
Architecture	Goes directly to the Architecture gallery
People & Culture	Goes directly to the People & Culture gallery

4.2 Places Navigation (Three Levels Deep)

Places has the most levels. Here is the full path:

1	Click Places → see two cards: National and International
2	Click National → see State cards (Karnataka, Kerala, Tamil Nadu...)
3	Click a State → see City cards (Badami, Hampi, Aihole...)
4	Click a City → see the photo gallery for that city

International works the same way:

1	Click International → see Country cards (Canada...)
2	Click a Country → see City cards (Banff...)
3	Click a City → see the photo gallery

For International photos, enter the Country name in the "State / Country" field in the curator. The system understands this field means Country when you select Places/International.

4.3 Recently Added

A gold pulsing banner appears on the home page whenever photos have been added in the last 14 days. Clicking it shows all recently added photos in one gallery, each with category tags shown below the photo.

4.4 Admin Tag Editor (Right-Click on any Photo)

On the website, you can right-click any photo and choose Edit Tags. This lets you update the category, location, and remarks for that photo directly in the browser.

1	Right-click any photo on the website
2	Choose Edit Tags from the menu
3	Enter the admin password when prompted
4	Update any fields and click Save
5	Make sure patch_tags.py is running on your Mac for changes to be saved

⚠️ **Changes made via the web editor are not permanent until you run Claude_mohangraphy.py to redeploy.**

5. Travel Stories — Blog

Travel Stories is the blog section of your website. It shows a list of your articles on the Travel Stories page. Clicking any article title opens the full write-up. At the bottom of each article are buttons linking to the relevant photo collections.

5.1 How the Blog Works

All your blog posts are stored in one file: blog_posts.json. The deploy script reads this file and builds the blog pages automatically.

- One file — blog_posts.json — contains all articles
- Each article is one { } block inside the [ ] brackets
- After editing blog_posts.json, run Claude_mohangraphy.py to publish

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

5.2 Adding a New Blog Post

You have two options:

Option A — Use blog_editor.py (for short articles)

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/blog_editor.py
```

A series of Mac dialog boxes will appear asking for:

- Place name and Country
- Dates visited
- Post title
- Place tag (must match the city name in your photo tags)
- Summary — 1-2 sentences shown on the blog index list
- Article body — paste your full write-up here
- Collections links — which photo collections to link at the bottom

Option B — Edit blog_posts.json directly (for long articles)

For longer articles, it is easier to edit the file directly. Open it in VS Code or nano:

```
# Open in nano (built into Mac):
nano /Users/ncm/Pictures/Mohangraphy/Scripts/blog_posts.json

# Or open in VS Code (if installed):
code /Users/ncm/Pictures/Mohangraphy/Scripts/blog_posts.json
```

5.3 Blog Post Format — Complete Example

```
[
  {
    "id": "tadoba-2026",
    "title": "In Search of the Elusive Tiger @ Tadoba",
    "place": "Tadoba, Maharashtra",
    "country": "",
    "dates_visited": "March 2026",
    "place_tag": "Tadoba",
    "summary": "2 days, 5 safaris, one unforgettable sighting.",
    "collections_links": ["Places", "Nature"],
    "body": "First paragraph here.\n\nSecond paragraph here.\n\nThird paragraph."
  }
]
```

5.4 Writing the Body Text — Rules

What you type	What appears on the website
\n\n (two backslash-n)	Paragraph break — starts a new paragraph
\n (one backslash-n)	Line break within the same paragraph
## My Heading	Makes a bold sub-heading (start the paragraph with ##)
[photo: filename.jpg]	Inserts a photo inline in the article
[photo: filename.jpg | Caption text]	Inserts a photo with a caption below it

Example with headings and photos:

```
"body": "Opening paragraph of your article.\n\n
## Safari Day One\n\n
What happened on day one.\n\n
[photo: 2026-0324-Tadoba-NCM-9512.JPG | The tiger, finally spotted]\n\n
## The Drive Back\n\n
Final paragraph here."
```

⚠️ **The body field must be one continuous line in JSON — no actual line breaks inside the quotes. Use \n\n for paragraph breaks.**

5.5 Multiple Articles in One File

When you have more than one article, separate them with a comma:

```
[
  {
    "id": "tadoba-2026",
    "title": "Tadoba article...",
    "body": "..."
  },
  {
    "id": "megamalai-2025",
    "title": "Megamalai article...",
    "body": "..."
  }
]
```

✅ **The last article has no comma after its closing }. Every article before it does have a comma.**

5.6 Clearing All Blog Posts

To start fresh with no articles:

```
echo '[]' > /Users/ncm/Pictures/Mohangraphy/Scripts/blog_posts.json
```

Then redeploy:

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

6. Storage, Files and Backups

6.1 Three Copies of Every Photo

Your Mac stores three versions of every photo:

Folder	What it contains
Photos/	Your originals — full resolution. Never delete these.
Thumbs/	800px thumbnails — auto-generated. Safe to delete anytime.
Web/	2048px web copies — auto-generated. Safe to delete anytime.

✅ **Thumbs/ and Web/ are completely disposable. Delete them when your drive is full and they will be rebuilt on the next deploy.**

6.2 What Gets Published to GitHub

Your original full-resolution photos never leave your Mac. GitHub only receives:

- index.html — the entire website in one file
- Thumbs/ — 800px thumbnails
- Web/ — 2048px web copies
- CNAME — your domain name

6.3 Freeing Up Hard Drive Space

Step 1 — Delete the generated folders (safe):

```
rm -rf /Users/ncm/Pictures/Mohangraphy/Thumbs/*
rm -rf /Users/ncm/Pictures/Mohangraphy/Web/*
```

Step 2 — Add your new photos to the Photos/ folder

Step 3 — Redeploy (thumbnails are rebuilt automatically):

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

The first deploy after deleting both folders will be slow — every photo is regenerated from scratch. For 500 photos expect 10–20 minutes. Subsequent deploys are fast.

6.4 When Your Hard Drive Fills Up

Buy an external hard drive (4TB costs around ₹7,000–8,000). Move your Mohangraphy folder to it, then update one line in Claude_mohangraphy.py:

```
# Find this line near the top of Claude_mohangraphy.py:
ROOT_DIR = "/Users/ncm/Pictures/Mohangraphy"

# Change it to your external drive mount point:
ROOT_DIR = "/Volumes/MyDriveName/Mohangraphy"
```

Redeploy once and everything works exactly as before.

6.5 Storage Size Estimates

Item	Approximate size
Original RAW photo	25–40 MB each
Original JPG photo	8–15 MB each
Web copy (2048px)	1–3 MB each
Thumbnail (800px)	200–400 KB each
1,000 RAW photos (originals)	~30 GB on Mac
1,000 photos on GitHub	~2 GB (web copies only)

⚠️ **Always keep your originals backed up in two places. An external drive for working files and a second backup (iCloud or another drive). Hard drives do fail.**

7. The Deploy Script

7.1 How to Run

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

7.2 What Happens When You Run It

1	Reads content.json for site settings and credentials
2	Reads photo_metadata.json for all photo data
3	Reads blog_posts.json for Travel Stories
4	Cleans up dead entries (photos that have been moved or deleted)
5	Generates thumbnails and 2048px web copies for any new photos
6	Builds the complete index.html with all CSS and JavaScript embedded
7	Commits and pushes to GitHub
8	GitHub Actions fires automatically — sends email notifications to subscribers
9	Site is live at mohangraphy.com within 30 seconds

7.3 What the Output Looks Like

```
MOHANGRAPHY DEPLOY

Step 1 — Checking for deleted photos...
  ✅ photo_metadata.json is clean

Step 2 — Building site...
  OK blog_posts.json loaded — 1 post(s)
  Generating thumbnails + 2048px web copies (97 photos)...
  Thumbnails ready: 97  |  Web copies ready: 97
  BUILD COMPLETE

Step 3 — Deploying to GitHub...
  ✅ Committed: Deploy 2026-04-05 — 12 file(s) updated
  ✅ Pushed to GitHub successfully!
  🌐 Live in ~30 seconds at: https://www.mohangraphy.com
```

7.4 Site Settings — content.json

This file controls all the site credentials and settings. Open it with any text editor to update:

Setting	Purpose
site.contact_email	Email address shown on Get In Touch page
site.ga_measurement_id	Google Analytics ID (starts with G-)
site.supabase_url	Your Supabase project URL
site.supabase_anon_key	Supabase anonymous key for browser features

Setting	Purpose
site.admin_password	Password for right-click Edit Tags on the website
site.plausible_domain	Plausible analytics domain

8. Visitor-Facing Features

8.1 Hero Section

- Full-screen slideshow — rotates through Megamalai photos every 3 seconds
- LIGHT · MOMENT · STORY tagline with your name
- EXPLORE button scrolls down to the Collections grid
- Gold pulsing banner appears when photos have been added in the last 14 days
- Banner text: "N photo(s) added recently — view it/them"

8.2 Photo Modal

Clicking any photo opens a full-screen view with:

- Large photo on the left
- Photo title (remarks) and location on the right panel
- Like button — heart turns gold when liked, count stored in Supabase
- Request Quote button — opens an email to you pre-filled with the photo details
- Navigation arrows — or swipe on mobile

8.3 Stay in the Loop — Subscriber Form

- Name (optional) and email address (required)
- Email is validated before submission
- Subscriber saved to Supabase subscribers table
- Duplicate emails get a friendly "already subscribed" message
- Every deploy triggers an email to all subscribers automatically

8.4 Visit Counter

- Counts every unique visit to the site
- Shown in the footer as "· 1,234 visits"
- To prevent your own visits being counted, run this once in your browser:

Visit this URL in Safari or Chrome on each of your devices:

```
https://www.mohangraphy.com?owner=yes
```

You must do this on each device (Mac, iPhone, iPad) separately. If you clear browser data, you will need to do it again.

9. Email Notifications

9.1 How It Works

Every time you deploy the site and new photos are detected, an email is sent automatically to every subscriber. You do not need to do anything — it fires by itself.

1	You run Claude_mohangraphy.py — it pushes to GitHub
2	GitHub detects new photos in the Photos/ folder
3	GitHub Actions runs the notify.yml workflow
4	Workflow fetches the subscriber list from Supabase
5	Sends one email per subscriber via Resend
6	Email is sent from: notifications@mohangraphy.com

9.2 Exporting Your Subscriber List

To see all your subscribers, run:

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/list_subscribers.py
```

This creates a timestamped Excel file in your Scripts folder with all subscriber names and emails.

9.3 GitHub Secrets Required

These are stored securely in your GitHub repository settings and must not be changed unless credentials change:

Secret Name	Value
RESEND_API_KEY	API key from resend.com dashboard
SUPABASE_URL	Supabase project URL
SUPABASE_ANON_KEY	Supabase anonymous key

10. Troubleshooting

10.1 Common Issues and Fixes

Script gives "SyntaxError: invalid syntax" on line 1

The file was corrupted — a stray character was added when it opened in TextEdit. Fix it:

```
sed -i '' '1s/^[^i]//' /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```

To prevent this: always right-click downloaded .py files and choose Save Link As — never open them in TextEdit.

Blog post not showing on the website

The blog_posts.json file likely has a formatting error. Fix it by running:

```
python3 -c "import json; json.load(open('Scripts/blog_posts.json'))" && echo OK
```

If it prints an error with a line number, that is where the problem is. Common causes: a real line break inside the body text, or a missing comma between articles.

Photos not appearing in the correct section

Check that the categories in the curator match exactly. The most common mistake is selecting the wrong category — for example putting a bird photo under Nature/Landscapes instead of Nature/Birds.

International photos not appearing

Make sure you entered the Country name in the State / Country field in the curator, and selected Places/International as the category. The city should be in the City field.

"Recently Added" appearing on top of every gallery

This was a known bug that has been fixed in the current version. If you see it again, make sure you are running the latest Claude_mohangraphy.py.

curator.py gives "NameError: scan_photos is not defined"

The curator file is missing a function. Run this in Terminal to fix it:

```
python3 - << 'EOF'
with open('/Users/ncm/Pictures/Mohangraphy/Scripts/curator.py', 'r') as f:
    c = f.read()
if 'def scan_photos' not in c:
    fix = '\n\ndef scan_photos():\n    files = []\n    for root, _, names in
os.walk(PHOTOS_DIR):\n        for n in sorted(names):\n            if
n.lower().endswith((\'.jpg\', \'.jpeg\', \'.png\', \'.webp\')):\n
files.append(os.path.join(root, n))\n    return files\n'
    c = c.replace('def main():', fix + 'def main():')
    with open('/Users/ncm/Pictures/Mohangraphy/Scripts/curator.py', 'w') as f:
        f.write(c)
    print('Fixed!')
EOF
```

10.2 Safe Commands Reference

These commands are safe to run at any time:

Task	Command
Check if JSON is valid	python3 -c "import json; json.load(open('Scripts/photo_metadata.json'))" && echo OK

Task	Command
Delete generated files (safe)	rm -rf Thumbs/* Web/*
See all photos in database	python3 -c "import json; data=json.load(open('Scripts/photo_metadata.json')); print(len(data), 'photos')"
Full redeploy	python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py

11. Quick Reference Card

Commands You Use Every Day

Task	Terminal Command
Tag new photos	python3 /Users/ncm/Pictures/Mohangraphy/Scripts/curator.py
Deploy / publish site	python3 /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
Add/edit a blog post	python3 /Users/ncm/Pictures/Mohangraphy/Scripts/blog_editor.py
Export subscriber list	python3 /Users/ncm/Pictures/Mohangraphy/Scripts/list_subscribers.py
Free up disk space	rm -rf /Users/ncm/Pictures/Mohangraphy/Thumbs/* /Users/ncm/Pictures/Mohangraphy/Web/*
Edit site settings	nano /Users/ncm/Pictures/Mohangraphy/Scripts/content.json
Edit blog posts	nano /Users/ncm/Pictures/Mohangraphy/Scripts/blog_posts.json

Files You Must Never Delete

File	Why it matters
Scripts/Claude_mohangraphy.py	The entire website lives here
Scripts/curator.py	Your tagging tool
Scripts/content.json	Site credentials and settings
Scripts/photo_metadata.json	All your photo tags — years of tagging work
Scripts/blog_posts.json	All your blog articles
Photos/	Your original photos
.github/workflows/notify.yml	Email notification system

Files That Are Safe to Delete Anytime

File / Folder	Why it is safe
Thumbs/	Rebuilt automatically on next deploy
Web/	Rebuilt automatically on next deploy
index.html	Rebuilt automatically on next deploy